

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 10.2: Case Study 1 — Paper Review Skill

Zhigang Feng

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Topic 10.2 Agenda

1. **Case Study 1** — building a paper-review skill from scratch

Case Study 1

A reusable quality-control workflow for paper review

Case Study 1: Why This Becomes a Skill

Research job: produce a structured referee-style review for a quantitative economics paper with quote-level evidence and quality gates.

- You may run this workflow many times across papers, coauthors, and seminars
- Quality matters more than speed because a wrong criticism is costly
- The task has internal stages with clear dependencies
- That makes it a good candidate for a reusable skill rather than a one-off prompt

The Naive Baseline Fails Fast

Tempting prompt: “Review this paper.”

What usually comes back:

- generic praise and generic weaknesses
- no verbatim anchor to the paper
- no proof-checking discipline
- no calibration to quantitative macro standards
- no filter removing low-confidence comments

The failure is not just model quality. It is the absence of structure, specialization, and a final editorial gate.

What a High-Quality Output Should Look Like

Shortened review template

`## Overall Feedback`

- `1. Estimand is not fully pinned down in Section 2.3`
- `2. Proof of Theorem 2 needs one missing justification`
- `3. Numerical section needs a reproducibility note`

`## Detailed Comments`

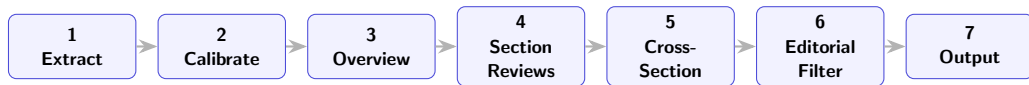
`### 1. Proof step in Theorem 2`

`Quote: "By dominated convergence ..."`

`Feedback: This step needs an explicit domination argument.`

Target properties: issue prioritization, exact anchoring, specific fixes, and a recommendation you can defend.

The 7-Phase Pipeline



- Parallelism happens inside phases, not across them
- The output of one phase becomes the acceptance criteria for the next
- This is a quality-control pipeline, not just a prompt chain

Why Decomposition Matters Here

Context

- long paper
- review rules
- output template

One call is too coarse.

Specialization

- proof review
- methodology review
- literature fit

One prompt does not do all well.

Quality

- generic comments
- invented quotes
- duplicated issues

You need a final filter.

Economics translation: this is like assigning specialized RA tasks, then forcing everything through one senior review before it goes out.

The Highest-Value Guardrails

1. **Confidence gate.** Do not assert an error unless the model can derive it step by step.
2. **Quote verification.** Comments that cite the paper must be anchored to real text.
3. **Editorial filter.** Remove generic advice, duplicates, and stylistic fluff.
4. **Humanization.** The final report should read like a serious economics referee report, not model boilerplate.

The lesson is broader than paper review: when the stakes are high, validation must be a named stage in the workflow.

Worked Example: Why the MPE Paper Is a Good Stress Test

Demo paper: *Markov Equilibria with Quasi-Hyperbolic Discounting*

- it mixes proofs, numerical work, and dynamic economic interpretation
- it forces the workflow to separate theorem checking from broader conceptual review
- it is hard enough that a one-shot prompt should fail

This is why case studies matter: you only learn the workflow once you see where the naive baseline actually breaks.

What the Economist Still Owns in Case Study 1

Agent side

- parse sections
- fan out specialized prompts
- draft comments
- enforce formatting

Economist side

- decide what counts as a serious objection
- reject false positives
- judge novelty and importance
- decide what feedback is worth sending

Skill logic: you are not outsourcing judgment. You are outsourcing disciplined first-pass execution.

When Should a Workflow Become a Skill?

Build a skill when ...	Stay ad hoc when ...
you will repeat the workflow many times	the task is one-off
the workflow has named validation stages	you are still exploring what the workflow should be
multiple users need consistent output	only one user needs the result
quality failures are expensive	output format is flexible

Case 1 verdict: this is a skill because repeatability and quality control are the point.